

User Continues
Conversation

Context Retrieval

Lookup Conversation ID

Check LRU Cache

Hit

Get from Cache

Miss

Cache Miss

Kafka Replay

Seek to Earliest Offset

Consume All Events

Reconstruct Full History

Message Count

<1000 msgs

Compression Decision

Total Tokens > Limit?
Default: 4000

>1000 msgs

No

Return Full History

Yes

Select Strategy

Entity

Hybrid
Default

Window

Full

Compression Strategies

Entity Graph
Preserve entities + key
messages

Hybrid
Window + Entity + Recent

Window Summary
Sliding window + LLM
summary

Full Summary
Single LLM summary

Compression Execution

Call LLM for Summarization

No
Retry

Preserve Key Elements:

